Temporally Stable Ultra-Compact Video Enhancement

Working manuscript: problem, protocol, and verified preliminary evidence

Internal research draft; authorship to be determined by the project team

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Abstract

Applying a compact image enhancer independently to video frames creates a quality–stability problem under a tight computation budget. We formulate the distinction between training-time temporal supervision and inference-time history, establish strong same-weight classical references, and document controlled evidence for adapting MobileIE. The completed experiments show that restoring the original expanded training parameterization improves a matched mixed-data baseline, while a fixed test-time local pooling conversion harms both evaluated domains. A completed factorial study of the root nonlinearity and a training-only temporal objective improves paired-real spatial quality but fails all three registered temporal/deployment screens. Its spatial and temporal effects must be separated. Matched learned-recursion controls also show negligible benefit of an internal feature-energy cue over an RGB gate, while exposing a quality–stability trade-off over fixed fusion weights. Completed fixed-budget adaptations of a published 696-parameter enhancer, a five-frame video denoiser, and a published deformation-consistency objective extend reference coverage while retaining training-history, convergence, and domain limitations. The fixed commutator adaptation reduces paired-real temporal error at a substantial spatial and perceptual cost; a fixed display-based metric further qualifies temporal-error gains. Common convolution budgets are reported without inferring phone speed. Completed native-resolution REDS development comparisons retain both seeds and show that recursion can reduce temporal error while harming some per-source spatial or perceptual scores. These findings do not establish a new video enhancement method. This internal draft contains no independent final-test claim or measured phone latency.

Draft status. This is a research manuscript under development, not a submission-ready paper. The final method, verified novelty, independent low-light video evaluation, and handset measurements remain incomplete. The numerical tables are generated from completed repository evidence. Training progress, partial checkpoints, and loss curves are excluded.

1 Problem and deployment constraints

An image enhancer maps a low-light observation x_t to an enhanced frame $z_t = f_\theta(x_t)$. A video system must also preserve legitimate scene changes, avoid noise-driven fluctuations, and fit its computation and memory budget. Low parameter count alone does not establish any of these properties. MobileIE [10] provides a useful compact starting point. Our 12-channel low-light deployment instance has 4,075 parameters; *4K parameters* does not mean a video resolution of 4K.

We distinguish a stateless predictor from a stateful one:

$$z_t = f_\theta(x_t), \qquad (z_t, h_t) = F_\theta(x_t, h_{t-1}).$$

A training objective may use neighbouring frames without introducing h_{t-1} at deployment. Conversely, a computation whose output changes for two histories with the same current frame cannot generally be represented exactly by a function of that current frame alone. Spatial convolution/BN folding does not eliminate this information dependency.

Our evaluation therefore reports quality, temporal error, parameters, convolution operations, online history, and measured pipeline latency as separate quantities. A small parameter change can still introduce full-resolution arithmetic or memory traffic. Inference motion estimation, sampling, padding, and buffering belong to the deployed pipeline even when they have no learned parameters. Server CPU measurements are not phone measurements.

1.1 The original nonlinear pathway

The retained MobileIE backbone consists of a convolutional head, a second feature block, channel/spatial gates, and an RGB readout [10]. Its feature self-transform (FST) has the form

$$q(u) = (w_1 u)(w_2 u) + b = au^2 + b,$$

where w_1, w_2 are shared scalars and b is channel-wise. Channel attention uses global average pooling followed by a 12×12 pointwise map and a sigmoid. The spatial gate depends on the channel-weighted feature maximum. The learned enhancement mapping, global statistics, and local nonlinear response can consequently interact.

For a scalar perturbation ϵ, the identity

$$q(u + \epsilon) - q(u) = 2au\epsilon + a\epsilon^2$$

motivates examining noise sensitivity at the root operator. If the perturbation has conditional mean zero, its conditional mean shift is $a\,\mathbb{E}[\epsilon^2]$. This elementary identity is not a new contribution. It neither proves that FST is the sole cause of video flicker nor establishes an unbiased correction for features after nonlinear convolutions, ISP processing, clipping, and quantization.

1.2 Known mechanisms and the remaining research question

Training-time reparameterization, channel multiplication, temporal regularization, and classical motion-compensated fusion are established tools. FastDVDnet [8], for example, learns multiframe denoising without an explicit flow estimator. TLC [2] addresses a different issue: the spatial-statistic mismatch between patch training and whole-image inference. Learnable safe rational activations also have clear precedents [6].

The unresolved question is whether an independently justified video adaptation can improve the quality–stability–cost trade-off over strong matched controls in this extremely compact setting. Neither assembling these known components nor obtaining a nonzero historical effect answers that question. The current candidate remains an experiment until its controlled results, external validity, and deployment costs support a defensible claim.

2 Related mechanisms and the scope of differentiation

Temporal training with image inference. StableLLVE [11] trains enhancement with synthetic motion and temporal consistency while its released U-Net predicts one frame at a time. Consequently, using temporal supervision without a deployed recurrent state already has direct LLVE precedent. The released checkpoint is explicitly trained on synthetic clean data and is qualified by its authors for noisy inputs. Our earlier transfer experiment preserves this limitation; it

does not replace common-data training or establish a fair ranking against a task-adapted compact model.

Explicit temporal information. FastDVDnet [8] and EMVD [4] illustrate two different deployment choices: a multiframe window and recurrent fusion. EMVD combines past observations recursively in a learned transform representation, then denoises and restores detail. Its real-RAW setting differs from the present RGB enhancement proxy. Neither a learned recurrent blend nor a small state alone distinguishes a new method. Our classical recursive reference tests the benefit of retained history, without claiming to reproduce EMVD.

Correspondence and decomposition. The 2026 VLLVE/VLLVE++ preprint [9] couples consistent decomposition with deep cross-frame interaction. Its described inference path uses a nearby reference frame; ablations also consider operation without a reference. Correspondence refinement operates during training. Removing that estimator alone does not remove the enhancement network's reference dependence. The deployed information and operators must therefore be stated explicitly. A training-only component does not establish single-frame inference.

The present draft differentiates experimental questions and information budgets, not an established novel method. A publishable contribution still requires a specific mechanism with controlled gains, stronger external validation, and measured deployment cost beyond these known ingredients.

3 Candidate and controlled questions

The following candidate was tested in the registered R44 study. Its complete results appear in Section 5.4; the registered temporal/deployment screens failed. No independent contribution is claimed.

3.1 A constrained root nonlinearity

We investigate the restricted rational response

$$q_\kappa(u) = \frac{au^2}{1+\kappa(u/s)^2} + b, \qquad s > 0, \quad \kappa \geq 0.$$

Each of the two FSTs receives one additional scalar κ. This is a constrained member of the known rational-activation family [6]; we do not claim the rational form, safe denominator, or learnable activation as a new invention. At $\kappa = 0$, the original response is recovered. For positive κ,

$$q'_\kappa(u) = \frac{2au}{[1+\kappa(u/s)^2]^2}, \qquad \sup_u |q'_\kappa(u)| = \frac{9|a|s}{8\sqrt{3\kappa}}.$$

The maximum occurs at $|u| = s/\sqrt{3\kappa}$. This scalar derivative bound does not establish a measured whole-network robustness or flicker bound; it also diverges when κ approaches zero.

The fixed scale s is the pre-FST feature RMS from eight predetermined training windows, computed separately for each layer and training seed. Calibration uses only training inputs and discards the temporary BN states. It fixes optimization units rather than selecting a scale on development quality. The learned denominator remains in deployment. The resulting predictor has 4,077 parameters, no temporal state in its raw path, and extra pointwise arithmetic compared with the 4,075-parameter square predictor. Only the spatial MBR convolutions and fixed-statistics BN are folded.

3.2 Training-only target-change compensation

Let W_t gather the previous frame into current-frame coordinates and let m_t be its validity mask. The candidate temporal objective is

$$\mathcal{L}_{\text{time}} = \frac{\sum_{t,p} m_t(p) \, \|[z_t - W_t z_{t-1}](p) - [y_t - W_t y_{t-1}](p)\|_2^2}{3 \sum_{t,p} m_t(p)}.$$

It is defined as zero when no points are valid. The frame reconstruction term still updates in that case. Subtracting the target change allows legitimate target motion or brightness change rather than assuming that the desired video is constant. Correspondences are estimated from training targets with the fixed half-resolution forward–backward estimator; they are not motion truth. Both predicted frames receive gradients. Deployment does not call this training correspondence estimator.

3.3 Factorial control

The fixed design crosses original square versus constrained rational FST with frame MSE versus $\mathcal{L}_{\text{MSE}} + 0.25\mathcal{L}_{\text{time}}$. All four models start from the same completed expanded-training checkpoint for each seed, with the same BN statistics. Adam is restarted for every arm. Each final model inherits 4,000 updates and receives 4,000 additional updates on the same prescribed windows. This controls for extra training, the objective, and the structural change; equal image/update budgets do not imply equal training FLOPs.

The primary structural screen compares the rational temporal model with the square model under the same temporal objective. A separate stateless deployment screen compares its raw output with both square controls' recursive postprocessing. A third screen compares all models with recursive postprocessing. A positive third screen cannot be reported as a stateless or flow-free result. Each registered screen includes both domains, both seeds, quality preservation, a temporal-error improvement requirement, and a worst-condition quality bound.

4 Experimental protocol and comparison scope

4.1 Development data

The current synthetic training pool contains 57 DAVIS training videos, excluding the three development sources *bear*, *lucia*, and *walking*. The synthetic development pilot uses 64 frames per source, at 480-pixel height, with two independently sampled noise conditions. There are three source videos and six conditions, not six independent videos. For a target RGB value y in $[0, 1]$, the degradation proxy is

$$v = 0.35y^{2.2}, \qquad x = Q_8(\text{clip}\,[\text{Poisson}(Pv)/P + \eta, 0, 1]), \quad \eta \sim \mathcal{N}(0, 0.008^2),$$

with $P \in \{20, 80\}$. Here Q_8 denotes conversion to the eight-bit observation. The present pilot adds no extra JPEG encoding, although the source frames are JPEG images. This is a controlled synthetic proxy, not a calibrated model of arbitrary RAW sensors or phone ISP video.

The fixed mixed-data study replaces half of each training batch with native crops from 12 verified paired ND-filter video sources in the project's OLiVES snapshot. Its four already-used paired-real development sources are videos 15, 47, 91, and 121, each evaluated on 16 frames with a 288×512 crop. Training and development source IDs are separated. Consecutive registered frame indices are available; independent sensor timestamps have not been verified. These daytime ND pairs must not be presented as natural nighttime validation.

Three original phone clips have additionally been inspected without synthetic degradation. They have no synchronized reference, so they support behavioural examples rather than reference PSNR or a perceptual flicker ranking. Independent final-test evaluation remains outstanding.

4.2 Quality and temporal measurements

Predictions are evaluated as floating-point RGB values after clipping to $[0, 255]$, before display rounding or video encoding. Sequence PSNR is the mean of full-frame RGB frame PSNR values,

$$\mathrm{PSNR}_{\mathrm{seq}} = \frac{1}{T} \sum_{t} 10 \log_{10} \frac{255^2}{\mathrm{MSE}_{\mathrm{RGB}}(z_t, y_t)}.$$

RGB SSIM uses an 11×11 Gaussian window with standard deviation 1.5 and the valid region obtained by excluding five boundary pixels. Edge MAE is the full-frame mean absolute error of horizontal and vertical luminance Sobel gradients, with the fixed gradient normalization used by the evaluation implementation.

For a fixed estimated reference flow and validity mask, temporal error is

$$E_{\mathrm{temp}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{\sum_{p,c} m_t^{\mathrm{ref}}(p) \left| [z_t - W_t^{\mathrm{ref}} z_{t-1}]_c(p) - [y_t - W_t^{\mathrm{ref}} y_{t-1}]_c(p) \right|}{3 \sum_p m_t^{\mathrm{ref}}(p)}.$$

Here $\mathcal{T}$ contains frame pairs with nonempty valid masks. The sequence summary first averages each valid frame pair, then averages the available pairs. The reference mask excludes an eight-pixel boundary. Reference flow is estimated from targets and held fixed across compared methods; it is not ground-truth motion. This metric is not a sufficient perceptual definition of flicker. Evaluation uses fixed reference fields and absolute RGB error. Training uses half-scale estimated fields, differentiable continuous bilinear gathering, and squared error. The two are not numerically identical.

Domains and training seeds are reported separately before aggregation. The present two-seed pilot does not establish statistical significance. We retain per-condition failures alongside mean improvements.

4.3 Strong same-weight references

For each spatial anchor, let r_t denote the raw prediction. The finite reference uses the previous raw output, while recursive fusion uses the previous fused output:

$$z_t^{\mathrm{fin}} = \tfrac{1}{2} r_t + \tfrac{1}{2} W_t r_{t-1}, \qquad z_t^{\mathrm{rec}} = \tfrac{1}{2} r_t + \tfrac{1}{2} W_t z_{t-1}^{\mathrm{rec}}.$$

The formulas apply only on accepted correspondences; invalid points and the first frame use the current raw output. Both estimate motion from their own adjacent raw predictions with the fixed half-resolution forward–backward estimator. Recursive fusion therefore also keeps the previous raw output. Neither reference requires a second enhancement CNN or a GT reset.

Published FastDVDnet weights [8] have been evaluated as fixed denoiser transfer, not as a retrained low-light video method. Its five-frame window uses two future frames. The documented pretraining source includes the three current DAVIS development IDs, so that transfer cannot establish unseen generalization. Same-domain retraining and broader published LLVE comparisons remain necessary.

Table 1: Completed anchor evidence: two-seed means on custom synthetic and paired ND development data. Historical rows differ in training; only the two R41 raw forms are a matched FP32 parameterization comparison.

Anchor/output	SYN PSNR	SYN E_{temp}	ND PSNR	ND E_{temp}
Dynamic synthetic (R34)	26.234	10.505	23.453	3.166
Fixed mixed Slim (R40)	25.134	11.363	24.861	3.628
Matched Slim FP32 (R41)	25.131	11.362	24.861	3.627
Expanded FP32 (R41)	26.186	10.007	25.847	3.630
Expanded + finite	26.394	7.286	25.734	2.804
Expanded + recursive	26.503	6.520	25.758	2.553
Earlier native training (R26)	–	–	27.138	3.313

4.4 Efficiency

Convolution MACs, pointwise work, history storage, and full pipeline time are distinct measurements. The original predictor has approximately 7.938 convolution GMAC at 1080p. This count omits non-convolution work. Online history counts use actual tensor storage rather than only logical channel slices. CPU timing is measured with two Torch threads and one OpenCV thread, and its workload and reference weights are stated explicitly. No handset latency is inferred from these measurements.

5 Verified preliminary evidence

The following results come from completed development experiments. They constrain method design and retain failed candidate screens. The source manifest and generated tables accompany this draft.

5.1 A stronger training baseline

Table 1 separates historical anchors from the matched training-form comparison. In the matched FP32 experiment, restoring MBR/BN training improves synthetic raw PSNR over direct Slim fine-tuning by 0.968 and 1.142 dB for the two seeds. The corresponding paired-real gains are 0.171 and 1.799 dB. Both deploy to the same 4,075-parameter spatial class. Training parameterization is therefore a material baseline choice. This is known spatial reparameterization, not a new video mechanism.

The expanded control uses 49,978 training parameters, including an inactive frozen warm head; 47,239 parameters are optimized. It is a controlled fine-tuning study, not a complete reproduction of the official warm-up, loss, and long-schedule training. To preserve the pretrained numerical starting function, the wide raw convolution executes its preserved and additional output blocks separately. This retains its parameters and mathematical operator but differs from the upstream single wide-kernel call.

The domain trade-off remains important. Expanded training approximately recovers the synthetic quality of the dynamic-synthetic anchor while improving its paired-real mapping. It still trails the earlier native training reference in paired-real PSNR. Those historical training schemes differ in several respects and do not isolate one causal factor. No single online model is obtained by taking the best weight separately for each domain.

Mean improvement does not imply reliable per-clip improvement. On paired-real video 121,

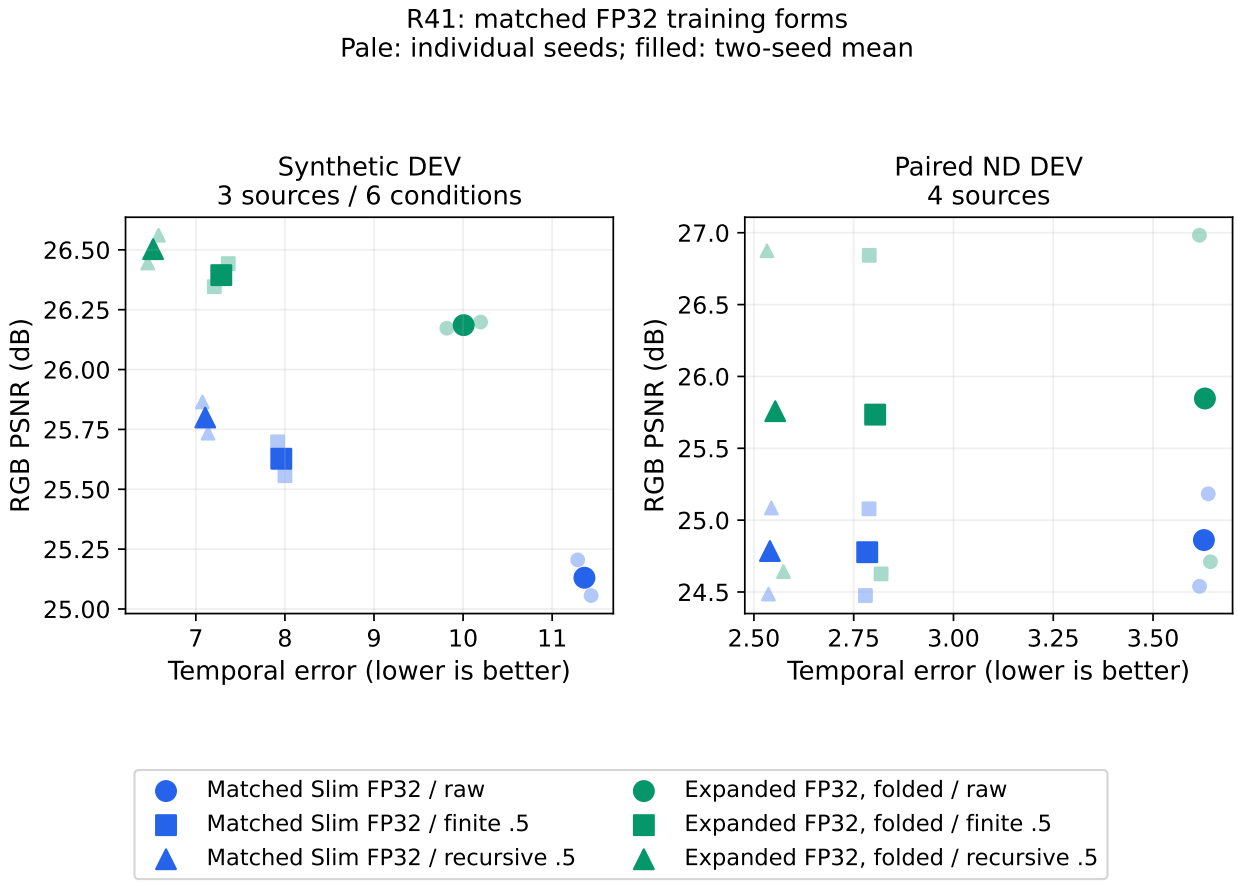


Figure 1: Completed matched FP32 training-form comparison. Pale points retain both individual seeds; solid points are their means. The real-domain spread and the quality–temporal trade-off are visible. Postprocessed outputs use motion and history; they are not stateless predictions.

Table 2: Classical references on the same R34 weights. Quality uses the full synthetic pilot. Resident CPU timing uses only the first 16 bear/P20 frames at 480×854, two Torch threads and one OpenCV thread. Ranges are run-average extrema, not confidence intervals or phone timings.

Output	PSNR	E_{temp}	History MiB	CPU ms/frame [range]
Raw	26.234	10.505	0.000	100.9 [80.9, 106.5]
Finite .5	26.466	7.543	4.691	184.1 [168.1, 188.5]
Recursive .5	26.610	6.735	9.382	186.5 [179.6, 192.5]

Table 3: Fixed known TLC192 versus the same R41 expanded weights. No retraining or window search.

Domain	Seed	Raw ΔPSNR	Raw temporal ratio
Synthetic	0	-0.305	1.0114
Synthetic	1	-0.415	1.0082
Paired ND	0	-2.312	1.0416
Paired ND	1	-2.343	1.0268

expanded seed 0 loses 2.606 dB against its matched Slim control and shows sky colour bias in the fixed-frame inspection. Its four-source real mean is 24.710 dB, versus 26.983 dB for expanded seed 1. Raw paired-real temporal error is not consistently reduced. All source/seed pairs and all four metrics remain part of the record.

5.2 Classical recursion is a necessary reference

On the same strong dynamic-synthetic anchor, recursive fusion improves the finite reference's quality and temporal error (Table 2). Its persistent fused output is a meaningful information advantage, not a negligible implementation detail. The recursive reference cannot be replaced by a weaker two-frame average when judging a proposed temporal model.

The measured CPU ranges overlap, so the small difference between the finite and recursive medians is not established as a significant speed difference. These timings concern the stated R34 weights and workload; they are not measurements of the later expanded anchor or the pending rational model. The extra stored RGB history and motion work remain part of deployment.

5.3 Fixed local statistics do not rescue this anchor

The fixed TLC [2] control replaces only the channel-pooling node with exact local averaging at a window of 192 pixels. Both development domains and both seeds lose raw PSNR; the paired-real temporal ratios also exceed the registered 1.02 upper bound (Table 3). Across the 20 raw seed–condition pairs, five PSNR deltas are positive, but all eight paired-real deltas are negative. The worst paired-real condition loses 3.716 dB.

The fixed instance and its local window search are stopped. This result does not prove that all local aggregation or global-statistic adaptations are ineffective. It also does not justify a speed claim: localizing the channel map adds approximately 0.299 convolution GMAC at 1080p, with prefix sums and memory traffic additional to that count.

Table 4: Complete R44 factorial, both-seed means. All models inherit the same R41 training form and add 4,000 updates. Raw inference is single-frame; recursive outputs retain motion and history. Rational denominators remain at inference.

Model	Output	SYN PSNR	SYN E_{temp}	ND PSNR	ND E_{temp}
Square/MSE	Raw	26.466	9.900	27.128	3.653
Square/temporal	Raw	26.463	9.328	27.084	3.482
Rational/MSE	Raw	26.448	9.892	27.586	3.660
Rational/temporal	Raw	26.428	9.315	27.547	3.484
Square/MSE	Recursive	26.707	6.470	27.108	2.570
Square/temporal	Recursive	26.526	6.257	27.018	2.508
Rational/MSE	Recursive	26.694	6.461	27.544	2.566
Rational/temporal	Recursive	26.495	6.250	27.459	2.504

5.4 Complete root-operator and temporal-objective factorial

Both seeds and all four arms complete their prescribed additional 4,000 updates. All final export checks pass, with maximum discrepancy 2.51×10^{-6}. Common CPU evaluation completes 240 sequence rows and 80 neural predictions of sequences. No intermediate checkpoint is used. Table 4 retains all raw and recursive arm means; finite outputs, individual seeds, and all condition comparisons remain in the full record.

The spatial effect is different from the temporal effect. Under MSE, the rational operator improves paired-real PSNR over its square control by 0.441 and 0.475 dB, with synthetic changes of -0.027 and -0.009 dB. Under the same temporal objective, paired-real gains are 0.461 and 0.465 dB, but temporal ratios range from 0.9925 to 1.0070 across the four domain–seed comparisons. Thus the primary structural screen preserves quality but fails the required temporal ratio of at most 0.90.

Temporal supervision reduces square-model raw temporal error by about 5.8% on synthetic data and 4.7% on paired-real data. The paired-real seed-1 PSNR change is -0.079 dB, outside the -0.05 dB preservation bound. The rational model shows a similar objective trade-off. These are finite regularization effects, not a demonstrated independent temporal advantage of the rational operator.

History remains useful. The rational temporal raw output has temporal error ratios of 1.435–1.444 on synthetic data and 1.352–1.359 on paired-real data against square-MSE recursion. It does not replace that strong history reference. With recursion applied to all models, the registered combination screen also fails: the synthetic mean loses about 0.212 dB against square-MSE recursion for only about 3.4% less temporal error. There is no new full-pipeline or phone-cost measurement for this candidate.

All three registered screens therefore fail. The local rational/temporal family, loss-weight, scale, learning-rate, training-length, and additional-seed search is stopped. The spatial quality signal is retained as a stronger reference, without a novelty claim. Continuing square-MSE training itself raises paired-real PSNR from about 25.85 dB at the earlier R41 checkpoint to 27.13 dB; that gain cannot be attributed to the new activation.

5.5 Published checkpoints and fixed-budget adaptations

Table 5 adds a 696-parameter published Nano enhancer [1] and a 2,479,096-parameter five-frame FastDVDnet [8]. Both complete two independent 8,000-update adaptations from their published weights using the same four supervised targets, crop coordinates, and augmentations as the spatial

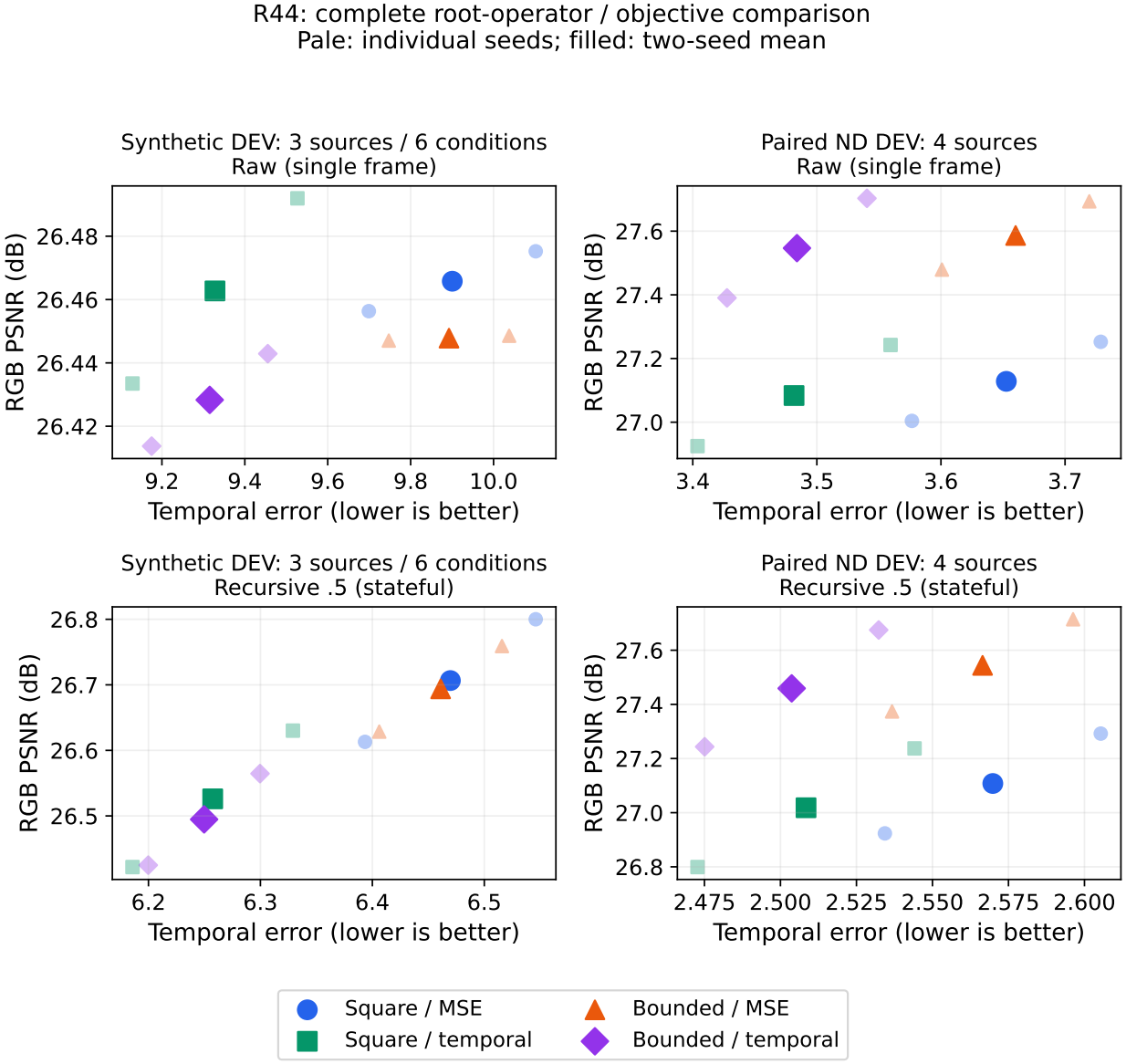


Figure 2: Complete R44 factorial. Both seeds and their means remain visible. Raw and recursive deployment paths are shown separately. The rational operator's paired-real quality gain does not imply an independent temporal gain; the temporal objective has a quality–stability trade-off.

Table 5: Completed published and adapted references on reused DEV data, all evaluated with the common CPU protocol. The original StableLLVE synthetic-clean and Nano LOL-v1 checkpoints have zero project updates and no claimed seed. Adapted backbones use 8,000 updates with the same four supervised targets; the RGB gate has 4,000 additional controller updates. FastDVDnet sees surrounding inputs and two future frames. Pretraining, training parameterization, noise conditions and compute differ; these are not matched-convergence architectural rankings. Current means five copies of the current input, a diagnostic. Recursion and RGB gating add motion/history cost beyond learned parameters.

Model/output	Seed	Params	SYN PSNR	SYN E_{temp}	ND PSNR	ND E_{temp}
StableLLVE clean raw	–	4316259	21.385	9.802	20.586	3.126
StableLLVE clean + rec.	–	4316259	21.541	6.668	20.539	2.476
Nano published raw	–	696	20.282	18.464	21.394	4.547
Nano published + rec.	–	696	21.933	10.454	21.387	2.964
Nano adapted raw	0	696	22.474	13.172	24.342	3.709
Nano adapted raw	1	696	22.204	13.131	23.902	3.708
Nano adapted + rec.	0	696	23.386	8.057	24.164	2.577
Nano adapted + rec.	1	696	23.033	8.039	23.743	2.581
FastDVDnet window	0	2479096	25.332	6.646	22.350	3.496
FastDVDnet window	1	2479096	22.582	7.553	22.964	3.530
FastDVDnet current	0	2479096	25.435	8.557	24.552	3.277
FastDVDnet current	1	2479096	23.997	8.526	24.872	3.316
MobileIE raw	0	4075	26.475	10.102	27.252	3.729
MobileIE raw	1	4075	26.456	9.699	27.004	3.577
MobileIE + rec.	0	4075	26.800	6.546	27.292	2.605
MobileIE + rec.	1	4075	26.613	6.393	26.923	2.534
MobileIE + RGB gate	0	4116	27.172	7.306	27.388	2.791
MobileIE + RGB gate	1	4116	27.052	6.976	27.080	2.677

reference. Only final weights are evaluated. This matches development updates and targets, not pretraining, training parameterization, computation, or full convergence. The Nano architecture and its original batch normalization are retained; four sequential batch-four forwards preserve that training semantics.

We additionally evaluate the original public StableLLVE synthetic-clean checkpoint [11], with 4,316,259 parameters and zero project updates, on the same inputs and CPU metrics. Its official BGR input convention and internal size handling are retained. Inference is frame-by-frame with no future input; the temporal training objective does not create an inference-time state. Classical recursion reduces temporal error, while slightly reducing paired-real PSNR. The released clean checkpoint does not reproduce the paper's noisy setting, and its transfer gap against task-adapted models is not a matched-training architectural ranking.

The Nano raw adaptation gains 1.922–2.192 dB on SYN and 2.508–2.949 dB on paired ND over its original LOL-v1 checkpoint. Its remaining gap to MobileIE in this budget does not establish a general architectural ranking. Applying the same classical recursion reduces its temporal error, but does not recover the paired-real quality gap.

FastDVDnet training supplies two additional source inputs on each side of the four targets. This avoids reflecting at every short sampled window, while adding observations; it is not an equal-information comparison to the earlier short-context control. There is no consistent improvement over that control across domains and seeds. On paired ND, true five-frame input loses 2.202 and 1.908 dB against five copies of the current frame, with temporal error about 6.7% and 6.5% higher. The latter is a same-weight diagnostic, not the published video configuration. The true window remains the video reference. Published AWGN pretraining overlaps three synthetic DEV sources; no unseen-source claim is made.

Table 6: Complete R45 comparison on the same frozen square-MSE base. Both-seed means; all fixed alpha points retained. All outputs except raw use motion and RGB history. Parameters omit unlearned motion work; no phone or pipeline cost is inferred.

Output	Params	SYN PSNR	SYN E_{temp}	ND PSNR	ND E_{temp}
Raw	4075	26.466	9.900	27.128	3.653
Fixed $\alpha = .1$	4075	26.733	8.922	27.188	3.316
Fixed $\alpha = .25$	4075	26.932	7.768	27.217	2.940
Fixed $\alpha = .5$	4075	26.707	6.470	27.108	2.570
Fixed $\alpha = .75$	4075	25.556	5.860	26.721	2.475
Fixed $\alpha = .9$	4075	23.833	5.936	26.147	2.575
Learned global	4076	26.922	7.893	27.218	2.979
RGB gate	4116	27.112	7.141	27.234	2.734
Internal cue	4116	27.116	7.122	27.232	2.741

Full phone inspection exposes a separate limitation. Both adapted Nano seeds retain purple stage colours but visibly over-brighten the scene; dark-region colour artifacts remain on the outdoor clip. The existing MobileIE reference also has severe exposure and colour failures there. These unpaired reused clips do not establish colour accuracy, temporal perception, or a phone quality ranking. These comparisons supply neither a new method, a reproduced standard LLVE benchmark, nor handset latency. The fixed reference budgets are closed rather than extended by local hyperparameter search.

5.6 Internal cues versus generic learned recursion

The next fixed experiment asks whether the compact network's internal response provides useful information beyond RGB for controlling history. Learned fusion has established precedent [4]; it is a reference mechanism here. The same square-MSE spatial checkpoint is frozen for each seed. A global scalar, a 41-parameter RGB gate, and a matched 41-parameter internal-cue gate each receive 4,000 updates using common four-frame windows and MSE. Own-raw motion, acceptance masks, and recursion are shared. All start from the classical $\alpha = .5$ function.

The internal inputs are two maps,

$$c(u) = \log(1 + 2|w_1 w_2|\,\mathrm{RMS}_C(u))\,,$$

read before the head and body FSTs in the same unchanged spatial forward pass. For a square FST, derivative RMS is simply a fixed rescaling of feature RMS. These maps are not an independent uncertainty theory or a calibrated output variance. The RGB gate replaces those two inputs with RGB chroma-deviation summaries while retaining the same capacity and six common RGB/motion summaries.

Both complete budgets and frozen-backbone checks pass. Final CPU evaluation finishes 180 rows and 20 CNN sequence calls; the latter capture cues and replay the earlier raw predictions with maximum discrepancy zero. Table 6 reports all five prescribed fixed weights and all learned controls. Neither domain nor seed is selected after evaluation.

The internal-cue result is nearly indistinguishable from the RGB control. Synthetic seed PSNR gains are only 0.0044 and 0.0027 dB; paired-real changes are -0.0017 and -0.0024 dB. Temporal ratios versus RGB are 0.9969–0.9978 on synthetic data and 1.0025–1.0031 on paired-real data. The registered independent-cue screen fails. Against fixed $\alpha = .5$, quality improves but temporal error increases by about 6–11%; that is not joint superiority under the registered criterion.

Generic learning remains a useful control. Both pixelwise gates choose accepted-pixel mean history weights around 0.34–0.36, compared with 0.228–0.236 for the learned global scalar. They are not jointly dominated by any of the five sampled fixed-weight points. This does not establish superiority over the continuous weight curve or an internal-cue contribution. Increasing fixed history weight to 0.9 lowers PSNR substantially and can also worsen temporal error. Motion error, resampling, occlusion, and bias remain possible causes; this experiment does not isolate one of them.

The feature-RMS controller's local architecture/input/objective/optimizer search is stopped. All controls are retained, and no candidate-specific pipeline or handset cost is inferred from this shared-computation quality experiment.

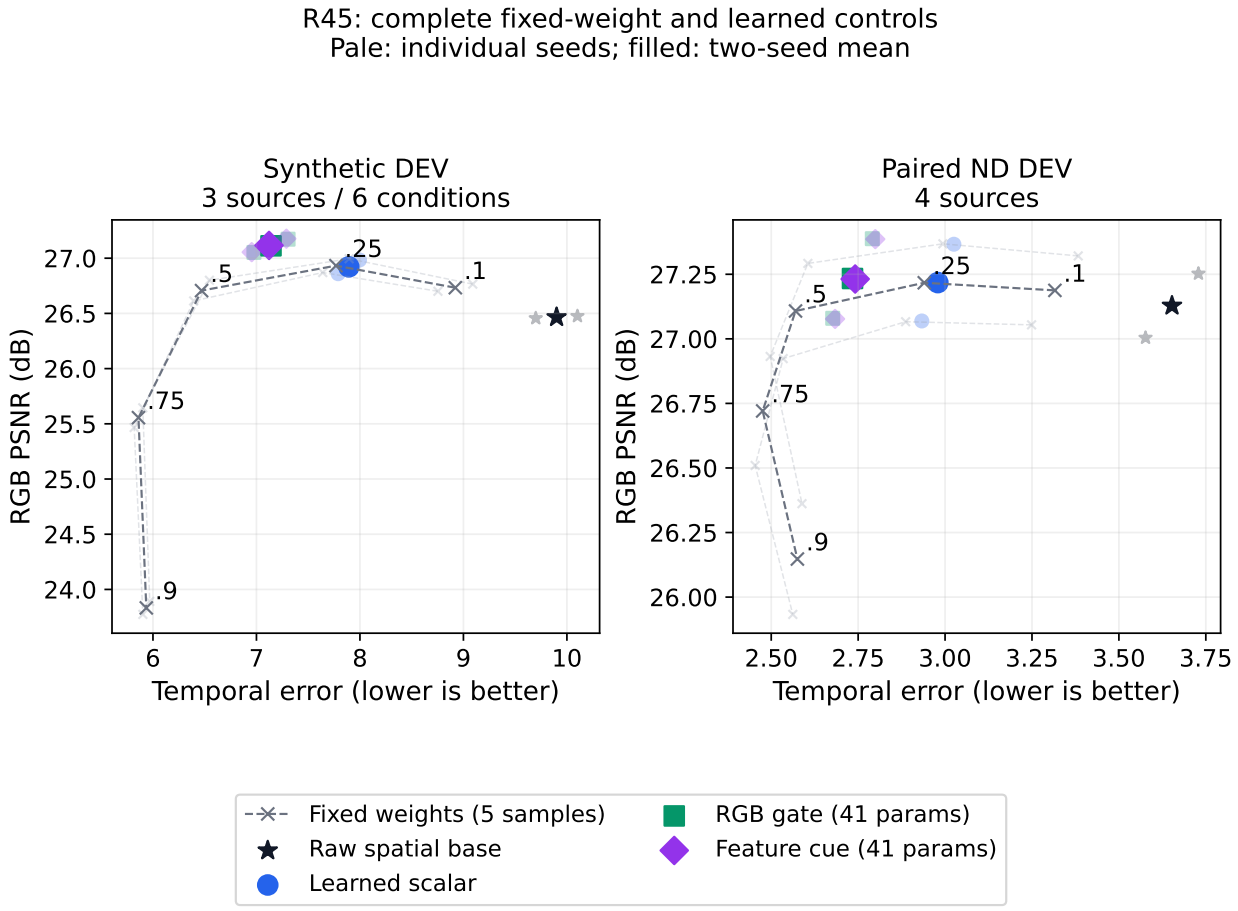


Figure 3: Complete R45 fixed-weight sample and learned controls. Pale traces and points retain both seeds. The RGB and internal-cue gates nearly overlap; their generic trade-off cannot be credited to an independent internal-cue mechanism. These are development results, not latency or final-test points.

6 Published-loss adaptation, perception and reference cost

A controlled published-loss reference. We adapt the official 4,316,259-parameter StableLLVE U-Net and deformation-consistency loss form [11] to the fixed mixed task. Both arms use the same public clean initialization, source windows, BGR ordering and 8,000 updates per seed. The

Table 7: Complete StableLLVE network/loss task adaptation. Every arm starts from the same public clean checkpoint and receives 8,000 updates per seed. The official L1 and weight-20 commutator form are retained, but training flow source, crop, initialization and schedule differ from the paper. Raw has no history/future input; rec. adds classical motion/history. JOD uses the fixed nominal display conditions in Table 9. This is neither a new method nor a full-paper reproduction.

Loss/output	Seed	SYN PSNR	SYN E_{temp}	SYN JOD	ND PSNR	ND E_{temp}	ND JOD
L1 raw	0	28.087	7.619	7.657	28.732	3.235	8.432
L1 raw	1	28.033	7.622	7.639	28.420	3.697	8.208
L1 + rec.	0	27.590	5.493	7.632	28.582	2.437	8.430
L1 + rec.	1	27.547	5.497	7.613	28.578	2.741	8.247
L1 + commute20 raw	0	25.368	7.743	6.992	25.853	2.905	7.765
L1 + commute20 raw	1	25.246	7.875	6.972	25.828	2.905	7.676
L1 + commute20 + rec.	0	25.204	5.789	6.912	25.632	2.353	7.651
L1 + commute20 + rec.	1	25.147	5.859	6.895	25.579	2.344	7.562

spatial arm minimizes L1; the second adds $20\|F(Wx) - WF(x)\|_1$, retaining both prediction paths' gradients. Here W uses training-GT motion as a deformation field applied to the same noisy source image. It does not create a separately noised adjacent observation. Compared with the paper, this changes CMP motion synthesis, crop size, initialization and training schedule. The original warp coordinate convention and CUDA backward nondeterminism are disclosed. This is a fixed-budget task adaptation, not a full reproduction, a convergence claim or a new method.

Table 7 contains only the four final 8,000-update models. Relative to its same-network spatial control, the commutator arm loses 2.59–2.88 dB raw PSNR and 0.53–0.67 predicted JOD across domains and seeds. SYN temporal error increases by 1.63–3.32%; paired-real error decreases by 10.19–21.44%, with lower spatial and perceptual quality. Applying the same recursive reference does not remove this trade-off. This fixed loss setting is stopped without a local weight, geometry, learning-rate, crop or training-length search. It does not invalidate the original paper or all equivariance objectives.

The L1 control is a useful stronger spatial reference: SYN raw PSNR is 28.03–28.09 dB, and its recursive temporal error is 5.49–5.50 versus 6.39–6.55 for the MobileIE recursive reference. However, its raw temporal error remains worse than that strong recursive reference. On paired-real seed 1, even its recursive error is 2.74 versus MobileIE's 2.53. Capacity, loss, pretraining and compute differ together; their separate effects cannot be inferred from this comparison. All 20 preassigned static contacts were inspected, including native center crops. These contacts are not full-video playback or human flicker judgments.

Matching the spatial objective does not close the gap. We separately train original MobileIE with spatial L1 under the same source-window schedule and two-stage 8,000-update budget as its existing MSE reference (Table 8). The original expanded training graph and initialization are retained; both deploy the same 4,075-parameter graph. The comparison concerns the full objective setting: different gradient scales interact with the shared gradient-clipping rule. Only the preassigned final weights are evaluated.

L1 does not give a consistent joint improvement. Raw temporal error falls by 1.18–3.59%, while paired-real raw PSNR falls by 0.37–1.03 dB and predicted JOD by 0.12–0.48. Recursive temporal error falls by only 0.22–1.80%, again with quality costs. SYN seed 1 raw gains just 0.036 dB while predicted JOD decreases slightly. This fixed reference is retained without an objective-mixture, learning-rate or seed search. All 20 prescribed contacts were inspected, without full-video playback or a human temporal study.

Table 8: Matched original MobileIE objective settings, final 8,000 updates only. Both objectives use the same source-window schedule, expanded training parameterization and two-stage Adam budget; L1/MSE gradient scale interacts with clipping. Every row deploys 4,075 parameters. Rec. adds the same classical motion/history reference. SYN comprises three source videos under two noise conditions; ND comprises four reused development clips. JOD uses the fixed assumptions in Table 9. No objective search, human study, handset or independent final-test claim.

Objective/output	Seed	SYN PSNR	SYN E_{temp}	SYN JOD	ND PSNR	ND E_{temp}	ND JOD
MSE raw	0	26.475	10.102	7.040	27.252	3.729	8.414
MSE raw	1	26.456	9.699	7.111	27.004	3.577	8.552
MSE + rec.	0	26.800	6.546	7.424	27.292	2.605	8.373
MSE + rec.	1	26.613	6.393	7.409	26.923	2.534	8.512
L1 raw	0	26.301	9.836	7.083	26.227	3.685	7.936
L1 raw	1	26.492	9.351	7.104	26.633	3.474	8.428
L1 + rec.	0	26.545	6.443	7.408	26.189	2.600	7.892
L1 + rec.	1	26.531	6.279	7.354	26.566	2.512	8.382

Table 9: Full-reference ColorVideoVDP v0.5.7 predictions for the existing references, with both adapted seeds. The original StableLLVE clean checkpoint has zero project updates and no training seed. Nominal 30 fps, original-sized SDR arrays, standard FHD display, 37.84 pixels/degree, 200 cd/m^2 peak and replicate temporal padding are assumed. These JOD predictions are not measured handset viewing, capture timing, human MOS or a separately calibrated flicker score.

Reference/output	Seed	SYN JOD	ND JOD
MobileIE raw	0	7.040	8.414
MobileIE raw	1	7.111	8.552
MobileIE + rec.	0	7.424	8.373
MobileIE + rec.	1	7.409	8.512
Nano raw	0	6.742	8.169
Nano raw	1	6.872	8.141
Nano + rec.	0	7.118	8.114
Nano + rec.	1	7.178	8.080
FastDVDnet window	0	7.432	7.680
FastDVDnet window	1	6.304	7.257
StableLLVE clean raw	–	6.262	6.802
StableLLVE clean + rec.	–	6.266	6.632

With both spatial references trained using L1, the U-Net still has 1.54–1.79 dB higher SYN raw PSNR and 1.79–2.51 dB higher paired-real raw PSNR. Thus the objective difference alone does not account for the observed gap, but architecture, capacity, pretraining and optimization remain coupled. This comparison does not identify capacity as the cause or establish a new temporal method.

Perceptual predictions qualify the temporal proxy. We evaluate ColorVideoVDP [5] on the complete cached sequences with one fixed display configuration (Table 9). The combined collection has 220 sequence-condition rows and 9,856 comparison frames, reusing 448 input observations; ten GT identity controls return exactly 10. These are repeated conditions, not independent video samples. In particular, the six SYN conditions are three source videos under two noise conditions. No enhancement network is rerun to obtain these scores.

For MobileIE on SYN, recursion improves predicted JOD by 0.30–0.38 alongside lower temporal error. On paired-real, the mean JOD changes are about −0.04 despite roughly 29–30% lower

Table 10: Reference deployment conditions at 1080 × 1920, batch one, per output frame. Conv GMAC counts only executed Conv2d multiply–accumulates; it omits bias, resizing, priors, pooling, activation, motion, warping and memory work. FastDVDnet uses the original uncached three first-stage and one second-stage calls. Rec. additionally retains two FP32 RGB frames plus motion/workspace; raw spatial networks have no temporal state. Identical StableLLVE graphs serve its public and adapted weights. No row is a measured handset latency or complete pipeline FLOP claim.

Network/output	Params	Conv GMAC	Future	Flow	Handset ms
MobileIE raw	4075	7.938	0	no	–
MobileIE + rec.	4075	7.938	0	yes	–
Nano raw	696	1.313	0	no	–
Nano + rec.	696	1.313	0	yes	–
FastDVDnet window	2479096	1319.208	2	no	–
StableLLVE raw	4316259	317.240	0	no	–
StableLLVE + rec.	4316259	317.240	0	yes	–

temporal error. The negative change is concentrated in video 15, while the other three sources have positive changes for both seeds. These small means are not evidence of statistically significant or human-perceived degradation. The spatially adapted U-Net also shows that lower temporal error need not improve JOD: on SYN its recursive JOD is about 0.026 below raw. Predicted perceptual quality, spatial fidelity and the temporal residual therefore need joint reporting. A dedicated human temporal study remains necessary; no display-condition grid is used to select favorable scores.

Efficiency remains a separate empirical requirement. The common Conv2d convention in Table 10 gives 317.24 GMAC for StableLLVE versus 7.94 for MobileIE at 1080p. This approximately forty-fold convolution-budget difference cannot be translated into a phone-latency ratio. The count excludes motion, warping, point operations and memory traffic, and the largest individual convolution output is not peak memory. Neither the 4,075-parameter starting point nor zero added inference cost is an imposed hard limit. A defensible deployment claim requires actual full-pipeline phone measurements for a candidate that first demonstrates joint quality and stability gains. No such new candidate is established by these references.

7 Fixed classical block-matching reference

We evaluate the original numerical core and default initialization of the IPOL VBM3D implementation [3] through an array interface, without changing its filtering or color-transform code. The spatial-only control disables temporal search ($N_f = 0$); the video control retains default temporal search ($N_f = 4$) on each full offline clip. Both receive one blind noise estimate from the first low-light frame: the median over RGB channels of centered Haar-HH MAD estimates, divided by 0.67448975 in the implementation's 8-bit amplitude units. No clean frame or known degradation parameter enters this estimate. One filtered input is reused by the two frozen 8,000-update MobileIE seeds. The same optional own-output motion-compensated recursive reference is then applied.

The two controls are deliberately limited: spatial-only VBM3D is not an optimized BM3D reproduction, and a full-clip video call is not a causal deployment. Its search radius alone does not certify an end-to-end future-wait bound across the filtering stages. An AWGN-oriented blind scalar is also not calibrated to the Poisson, clipping and quantization pipeline, or to a real camera ISP.

Table 11: Fixed original-core block-matching prefilters followed by the same frozen MobileIE weights. S-only disables temporal search in the VBM3D implementation; it is not an optimized BM3D reproduction. V-default uses the original video search on a full offline clip. A single blind first-frame noise estimate is shared by both modes. Rec. adds the same own-output motion/history reference. These reused development results and fixed-display JOD predictions do not establish calibrated noise, human judgments or a new method.

Prefilter/output	Seed	SYN PSNR	SYN E_{temp}	SYN JOD	ND PSNR	ND E_{temp}	ND JOD
None raw	0	26.475	10.102	7.040	27.252	3.729	8.414
None raw	1	26.456	9.699	7.111	27.004	3.577	8.552
None + rec.	0	26.800	6.546	7.424	27.292	2.605	8.373
None + rec.	1	26.613	6.393	7.409	26.923	2.534	8.512
S-only raw	0	24.351	8.142	6.917	26.689	3.525	8.383
S-only raw	1	24.204	7.778	6.959	26.479	3.380	8.530
S-only + rec.	0	24.278	5.772	6.931	26.725	2.540	8.328
S-only + rec.	1	24.043	5.646	6.923	26.411	2.474	8.475
V-default raw	0	23.978	7.554	6.781	26.605	3.466	8.380
V-default raw	1	23.807	7.233	6.816	26.394	3.327	8.527
V-default + rec.	0	23.878	5.707	6.749	26.641	2.522	8.324
V-default + rec.	1	23.637	5.599	6.735	26.328	2.459	8.471

Table 12: Descriptive CPU2 complete input-filter stages, including array I/O, in seconds per frame. Medians are over the registered sequence conditions; SYN has three source videos under two noise conditions. The same filtered input is reused across two enhancement seeds, so filtering is counted once. Shared-server timings have no isolated repeated latency trial and are not handset speed or an end-to-end future-wait bound.

Domain/size	Mode	Conditions	Median s/frame	Range
SYN 480×854	S-only	6	1.824	1.816–1.907
SYN 480×854	V-default	6	5.249	5.220–5.280
ND 288×512	S-only	4	0.677	0.658–0.695
ND 288×512	V-default	4	1.774	1.739–1.776

Enhancement weights are held fixed, so changes in their input distribution remain another possible cause; these causes have not been isolated.

All 16 domain–seed–filter–output means lower the temporal-error proxy but also lower PSNR and predicted JOD relative to the matching unfiltered enhancer output. On SYN, raw PSNR decreases by 2.124–2.649 dB and recursive PSNR by 2.522–2.976 dB. On paired-real ND, PSNR decreases by 0.513–0.651 dB and predicted JOD by 0.022–0.050. The latter small metric differences are not human preference or a significance claim. All 20 registered first/middle/last-frame contacts were actually inspected: grain reduction coexists with softer fur, stone and grass details; filtered recursive walking outputs show building-edge distortions. These sparse contacts neither replace full video inspection nor establish a causal mechanism or subjective score.

Table 12 retains descriptive complete filtering costs, including array I/O, from the registered shared-server CPU run with two CPU threads. Filtering is counted once across enhancement seeds. The measurements are not isolated repeated latency trials, handset measurements, full operation counts, or evidence for a 4K extrapolation. This fixed reference yields no robust joint quality and stability improvement. We retain it without searching noise estimates, windows, strengths or additional seeds, and keep the stronger unfiltered MobileIE recursive reference. This result neither rules out traditional denoising in general nor establishes a new method.

8 Cross-source development on native REDS frames

REDS was released for video deblurring and super-resolution [7]. We use its sharp validation source for a separate synthetic low-light development check, without changing any reference weights. Clips 000, 009, 019 and 029 and their first 64 frames were fixed before pixel acquisition. All 256 source frames retain their native 720×1280 resolution. The existing degradation applies a numerical sRGB power of 2.2, exposure factor 0.35, Poisson counts of 20 or 80, Gaussian standard deviation 0.008, clipping and uint8 quantization, with JPEG disabled. One registered PCG64 stream per source/condition is shared by all models. This is not a calibrated RAW/ISP simulator or the official REDS task.

Table 13: Cross-source synthetic development on four REDS sharp validation clips at native 720×1280, first 64 frames, with two fixed photon conditions. Every row averages the same four source videos; both adapted seeds are retained. Rec. adds the fixed classical motion/history reference to the corresponding raw output. ColorVideoVDP uses source-rate 24 fps and the same standard FHD display/replicate padding; these scores are not pooled with the 30 fps pilot tables. This is a custom low-light development protocol, not the official REDS task or independent final testing.

Network/output	Seed	20: PSNR	E_{temp}	JOD	80: PSNR	E_{temp}	JOD
Input	–	8.691	12.643	3.481	8.754	7.511	4.391
MobileIE MSE	0	25.809	11.777	5.781	28.458	6.961	7.422
MobileIE MSE	1	25.873	11.315	5.910	28.278	6.706	7.513
MobileIE MSE + rec.	0	27.662	6.544	6.701	28.843	4.418	7.762
MobileIE MSE + rec.	1	27.451	6.340	6.759	28.503	4.333	7.758
MobileIE L1	0	25.824	11.346	5.876	28.351	6.738	7.457
MobileIE L1	1	26.147	10.802	5.973	28.582	6.451	7.511
MobileIE L1 + rec.	0	27.438	6.353	6.740	28.624	4.344	7.735
MobileIE L1 + rec.	1	27.563	6.127	6.771	28.721	4.257	7.708
Nano	0	22.129	15.922	5.654	23.898	9.139	7.024
Nano	1	21.573	15.842	5.791	23.266	9.156	7.139
Nano + rec.	0	24.444	8.627	6.530	24.571	5.380	7.296
Nano + rec.	1	23.549	8.586	6.559	23.847	5.386	7.326
U-Net L1	0	28.265	7.894	6.920	30.562	5.512	8.031
U-Net L1	1	28.156	7.807	6.896	30.432	5.475	8.023
U-Net L1 + rec.	0	29.018	4.853	7.177	30.467	3.872	8.018
U-Net L1 + rec.	1	28.848	4.820	7.137	30.336	3.865	7.986
FastDVDnet window	0	25.361	6.418	6.701	25.960	4.905	7.599
FastDVDnet window	1	22.428	8.157	5.771	25.671	5.242	7.107

The five reference networks retain their completed 8,000-update adaptations and both seeds. There is no REDS fine-tuning or checkpoint selection. All 152 quality rows and 156 perceptual comparisons, including four unique GT controls scoring 10, complete normally. The four videos are the source units; two noise conditions and many frames do not add independent scenes. Source-rate 24 fps perceptual evaluation is kept separate from the earlier nominal 30 fps results.

MobileIE MSE recursion improves mean PSNR by 1.58–1.85 dB at 20 photons and 0.23–0.39 dB at 80 photons across the two seeds. Temporal-error ratios are 0.556–0.560 and 0.635–0.646, respectively. Yet two of its 16 source/condition/seed PSNR differences are negative, with the worst losing 0.284 dB; all 16 JOD differences are positive. Strong recursion remains necessary, without implying uniform per-source improvement.

The adapted spatial U-Net with recursion exceeds MobileIE MSE with recursion by 1.356–

1.833 dB and 0.228–0.476 JOD across these four condition/seed means. Its own recursion lowers temporal error in every pair, but loses PSNR in six and JOD in five of 16 pairs. The largest losses are 0.687 dB and 0.176 JOD. Lower temporal error therefore does not establish preserved image quality, even for a stronger spatial reference. Capacity, public pretraining and training computation are not matched by the common development updates.

Project adaptation uses the registered 57 DAVIS and 12 paired OLiVES training source IDs, with no REDS source IDs. That does not establish unseen public pretraining: author declarations and checkpoint/config lineages are available, but public per-image training manifests and a cross-source pixel audit are not. These four REDS sources have now been used for development. They cannot be relabelled as a blind final test, natural low-light evidence or a newly successful method.

9 Limitations and requirements for a complete paper

The present development pilot is small and repeatedly used. Synthetic noise, three DAVIS sources, and four paired daytime ND crops cannot establish general low-light video performance. The inspected phone clips show clipping and colour failures without synchronized ground truth. They do not replace a reference-based nighttime benchmark or a perceptual temporal study. Two training seeds expose variability but do not resolve it. The additional four native REDS sources broaden development coverage under two synthetic noise conditions, but retain public-pretraining uncertainty and do not supply natural nighttime ground truth or an independent final evaluation. Their source-rate 24 fps perceptual scores are kept separate from the original nominal 30 fps pilot.

Motion-compensated target-residual error helps separate desired changes from unwanted differences, but its estimated reference flow can fail and its valid mask excludes difficult regions. Spatial quality, per-condition failures, scene-change behaviour, and visual examples must accompany it. Lower temporal error alone is insufficient: an inappropriate enhancement mapping may also look numerically stable.

Published-weight transfers and fixed-budget task adaptations retain input-domain, convergence, training-compute and pretraining-overlap limitations. The completed Nano, source-context FastDVDnet and StableLLVE adaptations improve reference coverage but are not full standard LLVE reproductions. The fixed-display ColorVideoVDP predictions do not replace a human perceptual study, measured capture timing or handset viewing conditions. A conference comparison still requires appropriate published temporal enhancement methods under a documented common setting. The fixed classical references are strong controls, not full reproductions of all established video denoisers.

No candidate in this snapshot has a verified independent contribution or measured handset trade-off. A complete paper must establish the final method, explain its difference from relevant prior art, pass controlled structural/objective comparisons, measure the complete inference pipeline and state on a phone, and evaluate frozen decisions on genuinely independent data. The completed candidate's failure must guide a distinct hypothesis, with the registered local search stopped. Its spatial signal cannot be rewritten as temporal success. These requirements preserve the original research objective; the present manuscript is only the reusable foundation.

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